

## Department of Computer Science

### Modeling and Simulation Course

#### Code Assignment One

#### Assignment Instructions

Students will work **in groups of three (3)**. Each group is required to **compile, run, and analyze the provided simulation codes**, using the **supplied input data where available**, in order to deepen understanding of the modeling and simulation principles introduced in the initial lectures.

Each group shall prepare a **12-slide PowerPoint presentation** that addresses the following:

1. **Model Description**
  - Clearly explain **what system or process the code is modeling and/or simulating**.
  - Identify the **key components of the model**, such as inputs, state variables, outputs, and assumptions.
2. **Type of Simulation**
  - Classify the simulation
  - Justify your classification with reference to how the code operates.
3. **Simulation Experiments (Scenario Analysis)**
  - Design and run **at least two simulation experiments** by varying relevant parameters or input data.
  - Present and explain the **results and outputs** obtained from each scenario using tables, figures, or screenshots where appropriate.
4. **Model Evaluation and Improvement**
  - Identify **limitations, simplifications, or errors** in the code or model.
  - Propose **specific improvements or extensions** that would make the simulation more realistic, robust, or informative.
5. **Code Debugging and Corrections**
  - Some of the provided codes may contain **minor errors or inconsistencies**.
  - Identify any such issues, **correct them**, and briefly explain how the fixes affect the simulation results.

#### Submission Notes

- The presentation should be **clear, concise, and technically accurate**.
- Emphasis should be placed on **understanding and reasoning**, not just running the code.
- All group members are expected to contribute equally and be prepared to answer questions about any part of the work.